Teacher Wu Kuan | See It at a Glance #12 — Difference of Squares by Cut and Paste

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1. The trap question

Why does $(a + b)(a - b) = a^2 - b^2$?

2. The method

- Square a^2 minus corner square b^2 → L-shape with area $a^2 - b^2$.
- Cut and re-attach → rectangle $(a + b) \times (a - b)$; area unchanged.
- Use it for factoring and mental math.

3. Parent check method

Have your child cut the shape out of paper once — it sticks much better than the formula.

Practice

1. Use $a^2 - b^2$ to find $(7 + 3)(7 - 3)$.
2. $51 \times 49 = ?$
3. Factor $x^2 - 16$.
4. $102^2 - 98^2 = ?$
5. Factor $9x^2 - 1$.

Answers

1. 40
2. 2499
3. $(x + 4)(x - 4)$
4. 800 (200×4)
5. $(3x + 1)(3x - 1)$

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